

Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z5,ZL

Generated by MacAnalog-Symbolix

October 17, 2025

Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple **Z5 ZL**: $\frac{Z_5Z_Lg_m-Z_L}{Z_5g_m+2Z_Lg_m+1}$

$$H(z)=\frac{Z_5Z_Lg_m-Z_L}{Z_5g_m+2Z_Lg_m+1}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s)=\left(\infty,\infty,\infty,\infty,\frac{1}{C_5s},\frac{R_L}{C_LR_Ls+1}\right)$

$$H(s)=\frac{-C_5R_Ls+R_Lg_m}{C_5C_LR_Ls^2+g_m+s(2C_5R_Lg_m+C_5+C_LR_Lg_m)}$$

Parameters:

Q: $\frac{\sqrt{C_5}\sqrt{C_L}\sqrt{R_L}\sqrt{g_m}}{2C_5R_Lg_m+C_5+C_LR_Lg_m}$
wo: $\frac{\sqrt{g_m}}{\sqrt{C_5}\sqrt{C_L}\sqrt{R_L}}$
bandwidth: $\frac{2C_5R_Lg_m+C_5+C_LR_Lg_m}{C_5C_LR_L}$
K-LP: R_L
K-HP: 0
K-BP: $-\frac{C_5R_L}{2C_5R_Lg_m+C_5+C_LR_Lg_m}$
Qz: None
Wz: None

8.2 X-INVALID-NUMER-2 $Z(s)=\left(\infty,\infty,\infty,\infty,\frac{R_5}{C_5R_5s+1},\frac{1}{C_Ls}\right)$

$$H(s)=\frac{-C_5R_5s+R_5g_m-1}{C_5C_LR_5s^2+2g_m+s(2C_5R_5g_m+C_LR_5g_m+C_L)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_5}\sqrt{C_L}\sqrt{R_5}\sqrt{g_m}}{2C_5R_5g_m+C_LR_5g_m+C_L}$
wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_5}\sqrt{C_L}\sqrt{R_5}}$
bandwidth: $\frac{2C_5R_5g_m+C_LR_5g_m+C_L}{C_5C_LR_5}$
K-LP: $\frac{R_5g_m-1}{2g_m}$
K-HP: 0
K-BP: $-\frac{C_5R_5}{2C_5R_5g_m+C_LR_5g_m+C_L}$
Qz: None
Wz: None

8.3 X-INVALID-NUMER-3 $Z(s) = \left(\infty, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{-C_5 R_5 R_L s + R_5 R_L g_m - R_L}{C_5 C_L R_5 R_L s^2 + R_5 g_m + 2 R_L g_m + s (2 C_5 R_5 R_L g_m + C_5 R_5 + C_L R_5 R_L g_m + C_L R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_5} \sqrt{C_L} \sqrt{R_5} \sqrt{R_L} \sqrt{R_5 g_m + 2 R_L g_m + 1}}{2 C_5 R_5 R_L g_m + C_5 R_5 + C_L R_5 R_L g_m + C_L R_L}$
wo: $\frac{\sqrt{R_5 g_m + 2 R_L g_m + 1}}{\sqrt{C_5} \sqrt{C_L} \sqrt{R_5} \sqrt{R_L}}$
bandwidth: $\frac{2 C_5 R_5 R_L g_m + C_5 R_5 + C_L R_5 R_L g_m + C_L R_L}{C_5 C_L R_5 R_L}$
K-LP: $\frac{R_5 R_L g_m - R_L}{R_5 g_m + 2 R_L g_m + 1}$
K-HP: 0
K-BP: $-\frac{C_5 R_5 R_L}{2 C_5 R_5 R_L g_m + C_5 R_5 + C_L R_5 R_L g_m + C_L R_L}$
Qz: None
Wz: None

8.4 X-INVALID-NUMER-4 $Z(s) = \left(\infty, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_L g_m + s (C_5 R_5 R_L g_m - C_5 R_L)}{g_m + s^2 (C_5 C_L R_5 R_L g_m + C_5 C_L R_L) + s (C_5 R_5 g_m + 2 C_5 R_L g_m + C_5 + C_L R_L g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_5} \sqrt{C_L} R_5 \sqrt{R_L} g_m \sqrt{\frac{g_m}{R_5 g_m + 1}} + \sqrt{C_5} \sqrt{C_L} \sqrt{R_L} \sqrt{\frac{g_m}{R_5 g_m + 1}}}{C_5 R_5 g_m + 2 C_5 R_L g_m + C_5 + C_L R_L g_m}$
wo: $\sqrt{\frac{g_m}{C_5 C_L R_5 R_L g_m + C_5 C_L R_L}}$
bandwidth: $\frac{\sqrt{\frac{g_m}{C_5 C_L R_5 R_L g_m + C_5 C_L R_L}} (C_5 R_5 g_m + 2 C_5 R_L g_m + C_5 + C_L R_L g_m)}{\sqrt{C_5} \sqrt{C_L} R_5 \sqrt{R_L} g_m \sqrt{\frac{g_m}{R_5 g_m + 1}} + \sqrt{C_5} \sqrt{C_L} \sqrt{R_L} \sqrt{\frac{g_m}{R_5 g_m + 1}}}$
K-LP: R_L
K-HP: 0
K-BP: $\frac{C_5 R_5 R_L g_m - C_5 R_L}{C_5 R_5 g_m + 2 C_5 R_L g_m + C_5 + C_L R_L g_m}$
Qz: None
Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (\infty, \infty, \infty, \infty, R_5, R_L)$

$$H(s) = \frac{R_5 R_L g_m - R_L}{R_5 g_m + 2 R_L g_m + 1}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(\infty, \infty, \infty, \infty, R_5, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_5 g_m - 1}{2 g_m + s (C_L R_5 g_m + C_L)}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(\infty, \infty, \infty, \infty, R_5, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_5 R_L g_m - R_L}{R_5 g_m + 2 R_L g_m + s (C_L R_5 R_L g_m + C_L R_L) + 1}$$

$$\mathbf{9.4 \quad X-INVALID-ORDER-4} \quad Z(s) = \left(\infty, \infty, \infty, \infty, R_5, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_5 g_m + s (C_L R_5 R_L g_m - C_L R_L) - 1}{2g_m + s (C_L R_5 g_m + 2C_L R_L g_m + C_L)}$$

$$\mathbf{9.5 \quad X-INVALID-ORDER-5} \quad Z(s) = \left(\infty, \infty, \infty, \infty, \frac{1}{C_5 s}, R_L \right)$$

$$H(s) = \frac{-C_5 R_L s + R_L g_m}{g_m + s (2C_5 R_L g_m + C_5)}$$

$$\mathbf{9.6 \quad X-INVALID-ORDER-6} \quad Z(s) = \left(\infty, \infty, \infty, \infty, \frac{1}{C_5 s}, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 s + g_m}{C_5 C_L s^2 + s (2C_5 g_m + C_L g_m)}$$

$$\mathbf{9.7 \quad X-INVALID-ORDER-7} \quad Z(s) = \left(\infty, \infty, \infty, \infty, \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L R_L s^2 + g_m + s (-C_5 + C_L R_L g_m)}{s^2 (2C_5 C_L R_L g_m + C_5 C_L) + s (2C_5 g_m + C_L g_m)}$$

$$\mathbf{9.8 \quad X-INVALID-ORDER-8} \quad Z(s) = \left(\infty, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, R_L \right)$$

$$H(s) = \frac{-C_5 R_5 R_L s + R_5 R_L g_m - R_L}{R_5 g_m + 2R_L g_m + s (2C_5 R_5 R_L g_m + C_5 R_5) + 1}$$

$$\mathbf{9.9 \quad X-INVALID-ORDER-9} \quad Z(s) = \left(\infty, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, R_L \right)$$

$$H(s) = \frac{R_L g_m + s (C_5 R_5 R_L g_m - C_5 R_L)}{g_m + s (C_5 R_5 g_m + 2C_5 R_L g_m + C_5)}$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(\infty, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m + s (C_5 R_5 g_m - C_5)}{s^2 (C_5 C_L R_5 g_m + C_5 C_L) + s (2C_5 g_m + C_L g_m)}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(\infty, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m + s^2 (C_5 C_L R_5 R_L g_m - C_5 C_L R_L) + s (C_5 R_5 g_m - C_5 + C_L R_L g_m)}{s^2 (C_5 C_L R_5 g_m + 2C_5 C_L R_L g_m + C_5 C_L) + s (2C_5 g_m + C_L g_m)}$$

$$\mathbf{10 \quad X-INVALID-WZ}$$

10.1 X-INVALID-WZ-1 $Z(s) = \left(\infty, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_5 C_L R_5 R_L s^2 + R_5 g_m + s(-C_5 R_5 + C_L R_5 R_L g_m - C_L R_L) - 1}{2g_m + s^2(2C_5 C_L R_5 R_L g_m + C_5 C_L R_5) + s(2C_5 R_5 g_m + C_L R_5 g_m + 2C_L R_L g_m + C_L)}$$

Parameters:

Q: $\frac{2\sqrt{2}\sqrt{C_5}\sqrt{C_L}\sqrt{R_5}R_L g_m \sqrt{\frac{g_m}{2R_L g_m + 1}} + \sqrt{2}\sqrt{C_5}\sqrt{C_L}\sqrt{R_5}\sqrt{\frac{g_m}{2R_L g_m + 1}}}{2C_5 R_5 g_m + C_L R_5 g_m + 2C_L R_L g_m + C_L}$

wo: $\sqrt{2}\sqrt{\frac{g_m}{2C_5 C_L R_5 R_L g_m + C_5 C_L R_5}}$

bandwidth: $\frac{\sqrt{2}\sqrt{\frac{g_m}{2C_5 C_L R_5 R_L g_m + C_5 C_L R_5}}(2C_5 R_5 g_m + C_L R_5 g_m + 2C_L R_L g_m + C_L)}{2\sqrt{2}\sqrt{C_5}\sqrt{C_L}\sqrt{R_5}R_L g_m \sqrt{\frac{g_m}{2R_L g_m + 1}} + \sqrt{2}\sqrt{C_5}\sqrt{C_L}\sqrt{R_5}\sqrt{\frac{g_m}{2R_L g_m + 1}}}$

K-LP: $\frac{R_5 g_m - 1}{2g_m}$

K-HP: $-\frac{R_L}{2R_L g_m + 1}$

K-BP: $\frac{-C_5 R_5 + C_L R_5 R_L g_m - C_L R_L}{2C_5 R_5 g_m + C_L R_5 g_m + 2C_L R_L g_m + C_L}$

Qz: None

Wz: $\frac{\sqrt{-R_5 g_m + 1}}{\sqrt{C_5}\sqrt{C_L}\sqrt{R_5}\sqrt{R_L}}$

11 X-PolynomialError